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Research Statement

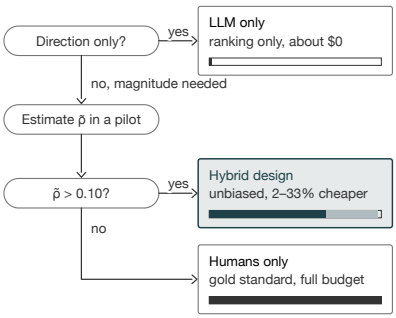
Overview

I trained in mathematics at TU Wien and the University of Zurich and wrote my PhD thesis at LMU Munich's Institute for Market-Based Management on simulating human decision-making with large language models. My papers measure how far answers written by a language model can replace human respondents, as in the pre-test of an ad, an offer or a price. They also study what the thresholds and grades of platforms and journals certify, from the salary band an employer-review platform displays to a journal's verdict on a manuscript.

● Measurement

When and How to Integrate AI Simulations into Behavioral Experimentation, in Revise & Resubmit at the **International Journal of Research in Marketing**, builds a benchmark of 70 treatment effects from 30 published marketing studies. I simulate each experiment with an open-weights language model at the level of the instrument: each synthetic respondent completes the original questionnaire in one randomized condition. Simulated and human effects correlate at $r = .73$, and 67.1 percent of the simulated effects fall inside the human 95 percent confidence interval. The calibration slope is .38, so the simulations understate effect sizes. They can rank ideas before a test is run; they cannot yet price them.

The paper turns this into a decision rule (right). If a study needs only the direction of an effect, the simulation alone gives a ranking at almost no cost. If it needs the size, a paired pilot estimates the correlation between human and simulated responses. Above 0.10, a hybrid design corrects the simulation with a human sample, stays unbiased and, in the paper's illustration, costs 2 to 33 percent less than humans alone. Below 0.10 the study runs on humans.



Synthetic Participants Reproduce Effects They Cannot Replace bounds that saving. The share of human respondents a study still needs falls as synthetic respondents are added, but it stops at $1 - \rho^2$, where ρ is the correlation within a study between a human's response and the simulator's response for that human. The floor holds however large the synthetic sample: at $\rho = 0.3$ the saving is 9 percent, and halving the human sample takes ρ of at least 0.71.

Post-training Makes Large Language Models Less Human-like, under review, introduces the Psych-201 dataset, to which I contributed as one of 79 authors, and measures behavioural alignment at scale. Post-training, the stage that turns a base model into an assistant, reduces alignment with human behaviour across model families and sizes, and conditioning a model on a participant's own information does not improve predictions for individuals.

In the many-analyst study forthcoming in the **International Journal of Research in Marketing**, 156 independent teams, mine among them, estimated brand price elasticities for 68 meat-substitute country-brand combinations from the same data. Across 10,415 estimates the average is −0.61, but the team averages range from −7.3 to +3.1. Choices on time aggregation, fixed effects, time controls and marketing-mix covariates explain the largest share of that spread.

■ Certification

After the Fax Machine Burns, submitted to **Management Science** as a Discussion Paper, starts from the cost of writing a manuscript, which language models have collapsed. It develops a system a publisher could run: every submission is published on arrival, and an open marketplace produces a public, revisable grade afterward. Modelling the five parties a publication system serves, the paper shows that authors, committees and readers weakly prefer a grade from the publisher's own panels to the

accept verdict.

Threshold Migration to the Patient, accepted at ANZAM 2026, follows a consumer device into a profession's gate: a wearable's algorithm now initiates triage while the clinician keeps disposition. Its difference-in-differences design uses the staggered international clearance of atrial-fibrillation detection to estimate the effect on specialist referrals.

kununu, the largest employer-review and salary platform in the German-speaking countries, shows a salary band only once enough employees have reported one. *The Platform Sets the Reference Wage*, which I am preparing, estimates with a regression discontinuity at that display threshold what the band does to applicant flows; a data partnership with kununu has been in negotiation since August. *The Matching Cost of Pay Transparency* models how firms respond to pay-transparency regulation and what follows for whom they hire; the formal model is complete and the estimation with Revelio Labs data is in progress.

Research at CBS

So far my benchmark compares simulations with published results. With CogLab and the experiments.cbs.dk participant pool I would run the simulated and the human version of the same study side by side on Danish samples, starting with pre-tests of ads and offers. SimuLabs, the open-source engine behind my simulation papers, runs the simulated arm.

Promotion planning rests on price elasticities, and the many-analyst study shows how far they depend on the analyst. With the data of a Danish retailer I would estimate how customers respond to prices and promotions, and report how the estimates move with the analytical choices the study identified.

Airbnb awards its Superhost badge to hosts with an average rating of at least 4.8 who also meet thresholds on response rate, cancellations and stays, and the public Inside Airbnb snapshots carry each listing's rating and badge. Hosts just above and just below the rating threshold differ little in quality but differ in the badge, so a regression discontinuity or a bunching design can separate what the badge certifies from what it adds. I would then study the ratings of Trustpilot, the review platform founded in Denmark and based in Copenhagen, as a certification of the same kind. Through CBS I could test the hiring question of the Matching Cost paper on Statistics Denmark's linked employer-employee registers.

I will publish the data and code of each paper when it is published.